Technical supplement

Traceable Entity Reconciliation with Gated Transaction Context: A Five-Dataset Evaluation

Companion to the detailed research manuscript and IEEE submission draft. Anonymous local review version; human authorship and declarations are pending. This supplement provides protocol details, complete configuration tables, result denominators, perturbation diagnostics and reproduction guidance.

The completed exploratory run contains four graph tasks and one offer-only benchmark. Its independent verification records 36 passing tests and 3,136 passing checks. WDC ERP transaction context and queries are not applicable. No human study was conducted.

Reading the evidence. Ambiguous F1 is conditional on a common candidate stratum. End-to-end pair F1 includes every original-entity cross-source positive. Final false merges are measured after one-to-one constraints. Stress accepted errors and coverage/risk are preconstraint pair metrics. These quantities are not interchangeable.

Machine-readable companions. generated/ contains all-seed query rows, all 45 main timing trials, full method/scenario means, official/disjoint WDC results, calibration summaries, scorer/canonicalization trials, source profiles and claim_registry.json. Exact trial-level study evidence remains in results/all-datasets-v1. Source hashes connect each manuscript table and figure to those measurements.

Contents. S1 population and protocol; S2 model/context semantics; S3 matching and accepted links; S4 paired effects; S5 official WDC cells; S6 disjoint WDC audit; S7 query definitions and denominators; S8 calibration; S9-S12 graph robustness; S13 WDC robustness; S14 scale; S15 timing; S16 reproduction/rights; S17 schema and evidence cases.

S1. Source populations and protocol

Full-source E1 imports and ER views are different scopes. Olist imports nine CSVs, DataCo the structured 180,519-row CSV, and BPI all 251,734 cases / 1,595,923 events. WWI full-source checks reuse its preserved verified core import. WDC checks the 27 pairwise split members without constructing transaction relationships.

Generic split seed is 20261012; validation-role seed is split seed + 101. Alias/training seeds are 17,29,43. Canonical groups are split before aliases. Selection, calibration and policy groups are disjoint; pairs connecting different roles are omitted. WWI retains its original split.

Dataset	Eligible	Selected	Train	Val.	Test
WWI	240	240	143	48	49
Olist	2913	2913	1747	583	583
DataCo	10075	1500	900	300	300
BPI2019	1378	1378	826	276	276

Olist uses all eligible natural repeated customers with two distinct observed accounts. DataCo selects 1,500 of 10,075 eligible customers by stable pre-alias hash. BPI selects all 1,378 eligible vendors. Both DataCo/BPI provide controlled source views with disjoint native histories, not independently annotated dirty-source identity pairs. Generic histories are capped at 40 transactions/cases per view.

WWI matching includes all purchase/supplier transactions in its preserved views and the first 5,000 sorted stock movements; the native E1 import has the complete in-scope movement table. Only three held-out supplier entities remain in WWI, limiting supplier-level conclusions.

E2 common pairs use original WWI token-prefix/category blocking and bounded K=32 on generic datasets. Ambiguous pairs have character similarity <0.95 or more than one near-name candidate (token-sort similarity >=0.65) for the left record. E5 selects K from 8,16,32 by validation recall target 0.95; posting cap 8K, 12 keys and two-million output guard. WWI chooses 8; generic choices 32 fail the validation recall target.

WDC trains nine hardness/development settings x three seeds = 27 model bundles, reusing each for 0/50/100% unseen tests: 81 evaluated cells. Each test contains 4,500 pairs and 500 positives. The used pairwise files have 10,422 offers / 1,600 product groups, not the whole benchmark corpus.

S2. Models, features and ablation semantics

The five-estimator catalog is equal across learned families: standardized class-balanced logistic regression C=0.1,1,10 (max_iter=1500), and random forests with 160 trees, depth 8/16, minimum leaf 2, class balancing and two jobs. Selection maximizes role-weighted F1 with its own threshold. Ties use catalog order.

Validation role weights restore full-validation prevalence. Sigmoid calibration uses logistic regression on logits clipped to [1e-6,1-1e-6], C=1e6, max_iter=1000. Policy selection maximizes weighted F1, with higher-threshold tie break. Reject probability is bounded by 0.10 and the accept threshold; no test-prior correction is performed.

Attributes have 20 features; WWI adds packaging equality/ratio/missingness for 23. Training-only TF-IDF and allowed text exclude truth, original master IDs, canonical keys, emails and passwords. Price currency agreement treats two empty strings as equal; no currency conversion is implemented.

Frozen embedding: sentence-transformers/all-MiniLM-L6-v2; revision 1110a243fdf4706b3f48f1d95db1a4f5529b4d41; onnx/model_quint8_avx2.onnx; batch 64; maximum 256 tokens; mean pooling with attention mask followed by L2 normalization. The 384-dimensional record embedding supplies one scalar pair cosine. It is not a fine-tuned pair model.

Generic context has 16 dimensions: category/activity cosine, degree similarity, distinct shared tags, weighted support, date mean/span, quantity mean/deviation, amount mean/deviation, date availability, quantity/amount compatibility, category contradiction and context availability. At least two projected tags and support one establish its gate; missing-value prefix tags can contribute.

WWI combines five native master context dimensions, 15 transaction dimensions and an independent inferred-anchor gate. At least two distinct exact-name/category-compatible anchors, excluding endpoints, and weighted support one are required. Packaging is excluded from anchor evidence. Purchased stock outers are converted to stock units; supplier outers and absent financial quantity remain separate.

Control	Interpretation
B4 / B5	Independent SQL extraction of the same permitted projection versus graph extraction; same model/calibration/policy, exact scores within 1e-12.
NoGate	Weighted learner without fallback, with its own selected policy; not a separately trained gate-free learner.
NoSupport	Retrained without support inputs, retaining the original final gate.
NoTransactions	WWI removes all 15 transaction dimensions; generic adapters remove date/numeric/unit dimensions only.
OneHop	Generic local transaction/activity evidence without category traversal; WWI additionally transforms native typed master neighbors.
UnitWeights	Unit-weight inputs to frozen weighted model and policy; not a separately retrained ablation.

S3. Conditional matching and final links

F1 below is the unweighted mean over three seeds on identical ambiguous candidate pairs. Differences between adapters are task differences, not a pooled ranking of datasets. BPI baseline fields are modal history descriptors; Olist has no customer names.

Method	WWI	Olist	DataCo	BPI
B0Exact	0.3402	0.7502	0.1994	0.0538
B1Fuzzy	0.7145	0.1039	0.2001	0.0639
B2Attributes	0.7688	0.6919	0.6962	0.1242
B3Embedding	0.4157	0.2327	0.2594	0.0840
B2EmbeddingAugmented	0.7626	0.7414	0.6962	0.1492
B4Flat	0.7888	0.7842	0.6750	0.5734
B5Graph	0.7888	0.7842	0.6750	0.5734
NoContext	0.7688	0.6919	0.6962	0.1242
NoGate	0.7763	0.7806	0.6710	0.5734
NoSupport	0.7879	0.7913	0.6780	0.5465
NoTransactions	0.7887	0.7474	0.6790	0.4264
OneHop	0.7688	0.6919	0.6946	0.4827
UnitWeights	0.7440	0.7842	0.6750	0.5734

Dataset	Accepted links	True accepted	False merges
WWI	107	106	1
Olist	1092	1055	37
DataCo	636	574	62
BPI2019	372	318	54

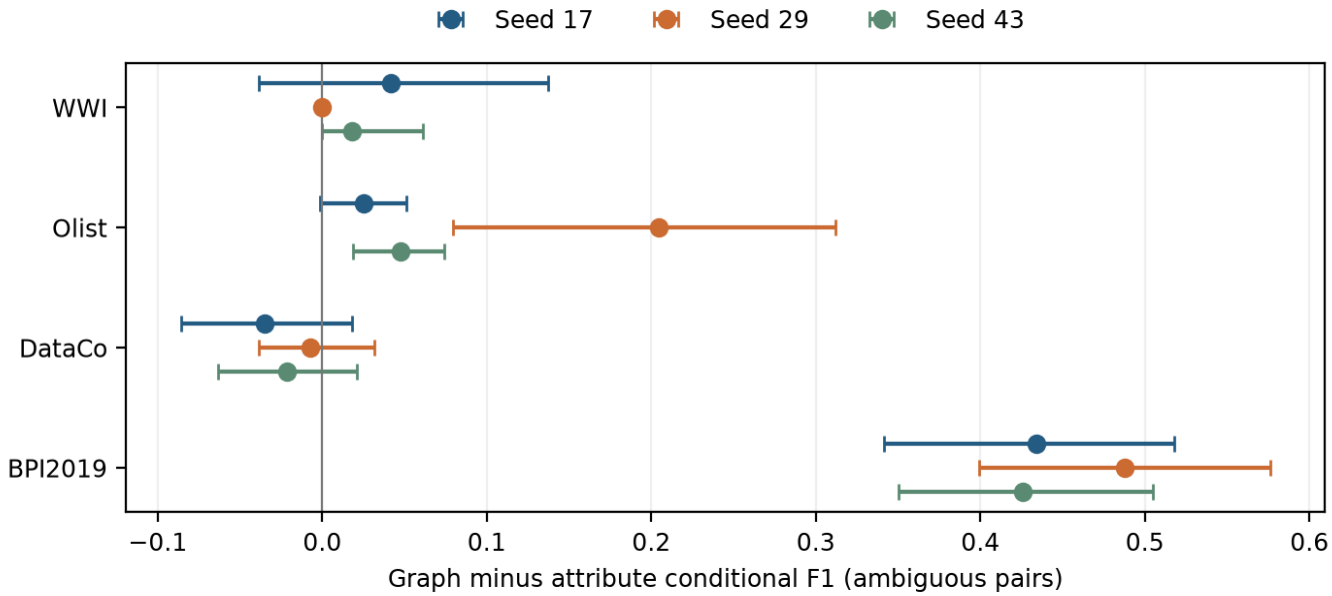
Final link totals sum trial outcomes over shared original populations, not independent production events. Candidate-level accuracy is evaluated before greedy constraints. Accepted false merges after constraints are distinct from stress false-positive rates.

S4. Paired effects and bootstrap specification

For each seed, resample canonical groups in the ambiguous candidate set 1,000 times with replacement. A same-group pair receives that group multiplicity m(g); a pair connecting distinct groups receives m(g)*m(h). The same weights apply to attributes and graph predictions. Report percentile endpoints 2.5/97.5 of graph-minus-attribute weighted F1.

These are explicitly defined endpoint-multiplicity group resamples, not independent pair samples. Three seeds share populations. Intervals are descriptive and exploratory, without a pooled interval or multiplicity-adjusted significance test.

Dataset	Seed	Delta F1	2.5%	97.5%
WWI	17	+0.0419	-0.0386	+0.1371
WWI	29	+0.0000	+0.0000	+0.0000
WWI	43	+0.0183	+0.0000	+0.0612
Olist	17	+0.0252	-0.0015	+0.0511
Olist	29	+0.2043	+0.0795	+0.3119
Olist	43	+0.0476	+0.0190	+0.0739
DataCo	17	-0.0347	-0.0860	+0.0182
DataCo	29	-0.0074	-0.0388	+0.0315
DataCo	43	-0.0212	-0.0631	+0.0213
BPI2019	17	+0.4341	+0.3416	+0.5177
BPI2019	29	+0.4876	+0.3993	+0.5764
BPI2019	43	+0.4261	+0.3500	+0.5049



S5. All official WDC pairwise cells

Every official pair is retained. F1 values are three-seed means. Nonzero training/test or validation/test offer intersections flag contamination; both must be checked. Hardness and unseen columns are percentages.

Hard	Size	Unseen	Attr.	Emb.	Attr+emb	Train/test	Val/test
20	small	0	0.529	0.357	0.570	0	4
20	small	50	0.570	0.388	0.611	97	105
20	small	100	0.570	0.390	0.600	0	0
20	medium	0	0.587	0.389	0.583	0	4
20	medium	50	0.603	0.411	0.620	155	105
20	medium	100	0.587	0.416	0.613	0	0
20	large	0	0.568	0.389	0.581	0	4
20	large	50	0.605	0.414	0.627	269	105
20	large	100	0.590	0.429	0.596	0	0
50	small	0	0.533	0.342	0.543	0	4
50	small	50	0.539	0.347	0.542	110	106
50	small	100	0.535	0.370	0.548	0	0
50	medium	0	0.555	0.381	0.564	0	4
50	medium	50	0.567	0.377	0.571	162	106
50	medium	100	0.538	0.417	0.555	0	0
50	large	0	0.558	0.381	0.572	0	4
50	large	50	0.588	0.377	0.607	274	106
50	large	100	0.538	0.417	0.578	0	0
80	small	0	0.446	0.323	0.450	0	4
80	small	50	0.490	0.336	0.490	110	103
80	small	100	0.470	0.347	0.482	0	0
80	medium	0	0.467	0.329	0.471	0	4
80	medium	50	0.519	0.369	0.529	159	103
80	medium	100	0.498	0.385	0.524	0	0
80	large	0	0.469	0.299	0.476	0	4
80	large	50	0.531	0.371	0.529	272	103
80	large	100	0.506	0.355	0.529	0	0

Eighteen cells have development/test offer overlap. Fully unseen cells have no such overlap. Hardness/test pools repeat across sizes, so configuration means are correlated descriptive summaries.

S6. WDC offer-disjoint diagnostics

The diagnostic subset keeps test pairs whose offers do not appear in training or validation. It changes denominators and is not a replacement official benchmark. Exact positive counts per seed/cell are retained in the source matching.csv.

Hard	Size	Unseen	Pairs	Attr.	Attr+emb
20	small	0	4469	0.531	0.573
20	small	50	2918	0.561	0.604
20	small	100	4500	0.570	0.600
20	medium	0	4469	0.588	0.584
20	medium	50	2510	0.588	0.611
20	medium	100	4500	0.587	0.613
20	large	0	4469	0.569	0.583
20	large	50	1889	0.591	0.610
20	large	100	4500	0.590	0.596
50	small	0	4465	0.536	0.545
50	small	50	2836	0.535	0.540
50	small	100	4500	0.535	0.548
50	medium	0	4465	0.557	0.566
50	medium	50	2498	0.542	0.551
50	medium	100	4500	0.538	0.555
50	large	0	4465	0.560	0.574
50	large	50	1946	0.560	0.586
50	large	100	4500	0.538	0.578
80	small	0	4476	0.446	0.451
80	small	50	2853	0.483	0.482
80	small	100	4500	0.470	0.482
80	medium	0	4476	0.468	0.471
80	medium	50	2564	0.487	0.508
80	medium	100	4500	0.498	0.524
80	large	0	4476	0.469	0.475
80	large	50	1998	0.513	0.516
80	large	100	4500	0.506	0.529

The manuscript headline WDC result uses the nine fully unseen official cells: attribute 0.5369, embedding-only 0.3919, augmented 0.5584. The mean augmented increment is 0.0215. No pooled confidence interval or test-based unseen-configuration selection is used.

S7. Query tasks, controls and denominators

Q1: matched source entities sharing projected tags. Q2: local positive-value transactions. Q3: B amount/count attributed to reconciled A. Q4: local entity/transaction/category paths. Q5: positive-value B transactions linked to A. Q6: cross-source identity pairs. Q7: B amount/count by tag attributed to A. Q8: lineage table for reconciled transaction paths.

Q2/Q4 are source-local. Other tasks depend on identities. SQL and graph answers agree exactly for all unreconciled, predicted and oracle maps; oracle answers are nonempty. Aggregates are cent-rounded source units. BPI amounts are initial anonymized case values; DataCo/Olist currency is unspecified. Multiple Q7 tag vocabularies overlap and are not an overall spend rollup.

Dataset	Query	Oracle answer range	Predicted mean F1
WWI	Q1	24-31	0.800
WWI	Q2	9446-9446	1.000
WWI	Q3	47-47	0.845
WWI	Q4	5047-8973	1.000
WWI	Q5	1067-1067	0.998
WWI	Q6	49-49	0.832
WWI	Q7	37-41	0.816
WWI	Q8	2298-2298	0.931
Olist	Q1	669-669	0.794
Olist	Q2	1408-1408	1.000
Olist	Q3	583-583	0.743
Olist	Q4	2816-2816	1.000
Olist	Q5	710-710	0.729
Olist	Q6	583-583	0.743
Olist	Q7	1179-1179	0.741
Olist	Q8	710-710	0.729
DataCo	Q1	1393-1393	0.739
DataCo	Q2	4558-4558	1.000
DataCo	Q3	300-300	0.747
DataCo	Q4	9116-9116	1.000
DataCo	Q5	2375-2375	0.742
DataCo	Q6	300-300	0.747
DataCo	Q7	2529-2529	0.745
DataCo	Q8	2375-2375	0.742
BPI2019	Q1	1019-1019	0.561
BPI2019	Q2	8588-8588	1.000
BPI2019	Q3	276-276	0.528
BPI2019	Q4	27366-27366	1.000
BPI2019	Q5	4308-4308	0.603
BPI2019	Q6	276-276	0.528
BPI2019	Q7	1082-1082	0.560
BPI2019	Q8	4574-4574	0.594

Ranges are over alias seeds; trial-level actual/expected/tp/precision/recall/F1 remain machine-readable. WWI additionally has 72 native query-control rows with purchasing/warehouse/finance semantics and native category/lineage Q7/Q8, separate from these adapted tasks.

S8. Calibration and decision reliability

Metrics use all clean candidate pairs, not only ambiguous pairs. Brier, log loss and ECE (ten equal-width bins) compare raw and sigmoid-calibrated scores. Values below average three seeds. Lower losses do not guarantee a correct canonical link or stable deployment probabilities.

Dataset	Method	Mode	Brier	Log loss	ECE
BPI2019	Attr.	calibrated	0.0251	0.1120	0.0021
BPI2019	Attr.	raw	0.1711	0.4903	0.3301
BPI2019	Graph	calibrated	0.0149	0.0541	0.0052
BPI2019	Graph	raw	0.0180	0.0725	0.0190
DataCo	Attr.	calibrated	0.0129	0.0407	0.0088
DataCo	Attr.	raw	0.0213	0.0777	0.0366
DataCo	Graph	calibrated	0.0138	0.0443	0.0073
DataCo	Graph	raw	0.0189	0.0678	0.0291
Olist	Attr.	calibrated	0.0211	0.0850	0.0111
Olist	Attr.	raw	0.0765	0.2297	0.1332
Olist	Graph	calibrated	0.0189	0.0779	0.0182
Olist	Graph	raw	0.0791	0.2372	0.1403
WWI	Attr.	calibrated	0.0077	0.0272	0.0078
WWI	Attr.	raw	0.0098	0.0418	0.0104
WWI	Graph	calibrated	0.0082	0.0300	0.0079
WWI	Graph	raw	0.0094	0.0420	0.0102

Coverage counts pair decisions accepted or rejected, before one-to-one constraints. Selective risk is error among those decided pairs. Accepted error rate is false positives among accepted candidate pairs. Abstention is the remainder. These rates depend on both acceptance and rejection policies and differ from the final-link audit.

Exact reliability bins and threshold coverage/risk curves are retained per graph seed and WDC configuration. Small, imbalanced calibration and policy roles introduce uncertainty; validation prevalence weighting does not correct unknown deployment priors. Comparisons of calibrated losses and ECE can disagree, as on Olist.

S9. WWI robustness on frozen pairs

Three-seed means under controlled additional defects. Candidates, estimators and thresholds are fixed; corrupted retrieval is not evaluated. The first table reports end-to-end pair F1 for selected controls. All 13 methods remain in robustness_means.csv and trial-level robustness artifacts.

NoTransactions and OneHop have adapter-specific meanings described in S2. UnitWeights retains the weighted estimator/policy. False-positive acceptance and coverage below are preconstraint pair diagnostics.

Condition	Attr	Graph	No gate	No sup	No tx	1 hop	Unit
Clean	0.797	0.826	0.809	0.820	0.826	0.797	0.794
No anchors/support	0.797	0.797	0.731	0.797	0.797	0.797	0.797
Attribute noise 30%	0.578	0.578	0.577	0.578	0.578	0.578	0.594
Attribute noise 50%	0.585	0.585	0.585	0.582	0.585	0.585	0.598
Optional missing 10%	0.795	0.821	0.784	0.822	0.830	0.795	0.798
Optional missing 30%	0.746	0.730	0.600	0.763	0.807	0.746	0.740
Optional missing 50%	0.733	0.696	0.523	0.746	0.803	0.733	0.716
Edges removed 15%	0.797	0.821	0.789	0.806	0.830	0.797	0.783
Edges removed 30%	0.797	0.794	0.770	0.792	0.801	0.797	0.771
Typed rewire 15%	0.797	0.794	0.560	0.783	0.817	0.797	0.756
Typed rewire 30%	0.797	0.788	0.571	0.793	0.789	0.797	0.783
Lineage missing 30%	0.797	0.797	0.715	0.797	0.797	0.797	0.794

Condition	Coverage	Selective risk	Accepted errors
Clean	0.982	0.003	0.034
No anchors/support	0.980	0.003	0.035
Attribute noise 30%	0.983	0.011	0.027
Attribute noise 50%	0.984	0.012	0.017
Optional missing 10%	0.983	0.004	0.033
Optional missing 30%	0.980	0.006	0.065
Optional missing 50%	0.982	0.007	0.088
Edges removed 15%	0.982	0.003	0.035
Edges removed 30%	0.979	0.003	0.028
Typed rewire 15%	0.981	0.005	0.057
Typed rewire 30%	0.979	0.003	0.036
Lineage missing 30%	0.980	0.003	0.035

Typed rewiring retains entity kind, silo and split. Full visible lineage does not establish correctness; missing prefix tags can still count toward generic support. Perturbation logs are evaluator-only. Source graphs are copied for stress rather than mutating the native imports.

S10. Olist robustness on frozen pairs

Three-seed means under controlled additional defects. Candidates, estimators and thresholds are fixed; corrupted retrieval is not evaluated. The first table reports end-to-end pair F1 for selected controls. All 13 methods remain in robustness_means.csv and trial-level robustness artifacts.

NoTransactions and OneHop have adapter-specific meanings described in S2. UnitWeights retains the weighted estimator/policy. False-positive acceptance and coverage below are preconstraint pair diagnostics.

Condition	Attr	Graph	No gate	No sup	No tx	1 hop	Unit
Clean	0.654	0.725	0.732	0.729	0.692	0.654	0.725
No anchors/support	0.654	0.688	0.690	0.688	0.688	0.654	0.688
Attribute noise 30%	0.497	0.550	0.576	0.550	0.523	0.497	0.550
Attribute noise 50%	0.451	0.492	0.522	0.492	0.473	0.451	0.492
Optional missing 10%	0.560	0.675	0.687	0.674	0.620	0.560	0.675
Optional missing 30%	0.394	0.551	0.597	0.552	0.448	0.394	0.551
Optional missing 50%	0.249	0.456	0.537	0.453	0.302	0.249	0.456
Edges removed 15%	0.654	0.715	0.604	0.719	0.690	0.654	0.715
Edges removed 30%	0.654	0.705	0.486	0.709	0.692	0.654	0.705
Typed rewire 15%	0.654	0.692	0.586	0.706	0.692	0.654	0.692
Typed rewire 30%	0.654	0.683	0.501	0.700	0.690	0.654	0.683
Lineage missing 30%	0.654	0.708	0.523	0.711	0.690	0.654	0.725

Condition	Coverage	Selective risk	Accepted errors
Clean	0.913	0.014	0.126
No anchors/support	0.916	0.016	0.109
Attribute noise 30%	0.940	0.024	0.128
Attribute noise 50%	0.946	0.026	0.131
Optional missing 10%	0.860	0.016	0.132
Optional missing 30%	0.739	0.020	0.168
Optional missing 50%	0.641	0.020	0.161
Edges removed 15%	0.914	0.015	0.120
Edges removed 30%	0.914	0.016	0.124
Typed rewire 15%	0.913	0.015	0.124
Typed rewire 30%	0.913	0.016	0.120
Lineage missing 30%	0.914	0.015	0.121

Typed rewiring retains entity kind, silo and split. Full visible lineage does not establish correctness; missing prefix tags can still count toward generic support. Perturbation logs are evaluator-only. Source graphs are copied for stress rather than mutating the native imports.

S11. DataCo robustness on frozen pairs

Three-seed means under controlled additional defects. Candidates, estimators and thresholds are fixed; corrupted retrieval is not evaluated. The first table reports end-to-end pair F1 for selected controls. All 13 methods remain in robustness_means.csv and trial-level robustness artifacts.

NoTransactions and OneHop have adapter-specific meanings described in S2. UnitWeights retains the weighted estimator/policy. False-positive acceptance and coverage below are preconstraint pair diagnostics.

Condition	Attr	Graph	No gate	No sup	No tx	1 hop	Unit
Clean	0.674	0.649	0.644	0.650	0.654	0.671	0.649
No anchors/support	0.674	0.661	0.631	0.657	0.666	0.671	0.661
Attribute noise 30%	0.510	0.488	0.489	0.487	0.493	0.508	0.488
Attribute noise 50%	0.442	0.414	0.416	0.413	0.424	0.436	0.414
Optional missing 10%	0.614	0.591	0.585	0.593	0.609	0.614	0.591
Optional missing 30%	0.470	0.439	0.435	0.433	0.492	0.471	0.439
Optional missing 50%	0.349	0.302	0.299	0.302	0.371	0.353	0.302
Edges removed 15%	0.674	0.644	0.643	0.648	0.653	0.669	0.644
Edges removed 30%	0.674	0.656	0.642	0.659	0.654	0.670	0.656
Typed rewire 15%	0.674	0.652	0.646	0.658	0.656	0.664	0.652
Typed rewire 30%	0.674	0.650	0.651	0.651	0.656	0.662	0.650
Lineage missing 30%	0.674	0.654	0.577	0.652	0.657	0.673	0.649

Condition	Coverage	Selective risk	Accepted errors
Clean	0.988	0.017	0.375
No anchors/support	0.991	0.017	0.382
Attribute noise 30%	0.991	0.022	0.399
Attribute noise 50%	0.991	0.025	0.442
Optional missing 10%	0.985	0.019	0.396
Optional missing 30%	0.980	0.023	0.491
Optional missing 50%	0.981	0.025	0.541
Edges removed 15%	0.988	0.017	0.380
Edges removed 30%	0.988	0.017	0.372
Typed rewire 15%	0.989	0.017	0.379
Typed rewire 30%	0.988	0.017	0.376
Lineage missing 30%	0.990	0.017	0.381

Typed rewiring retains entity kind, silo and split. Full visible lineage does not establish correctness; missing prefix tags can still count toward generic support. Perturbation logs are evaluator-only. Source graphs are copied for stress rather than mutating the native imports.

S12. BPI2019 robustness on frozen pairs

Three-seed means under controlled additional defects. Candidates, estimators and thresholds are fixed; corrupted retrieval is not evaluated. The first table reports end-to-end pair F1 for selected controls. All 13 methods remain in robustness_means.csv and trial-level robustness artifacts.

NoTransactions and OneHop have adapter-specific meanings described in S2. UnitWeights retains the weighted estimator/policy. False-positive acceptance and coverage below are preconstraint pair diagnostics.

Condition	Attr	Graph	No gate	No sup	No tx	1 hop	Unit
Clean	0.110	0.490	0.490	0.459	0.377	0.405	0.490
No anchors/support	0.110	0.007	0.000	0.000	0.007	0.405	0.007
Attribute noise 30%	0.084	0.465	0.465	0.381	0.335	0.342	0.465
Attribute noise 50%	0.074	0.449	0.449	0.350	0.317	0.322	0.449
Optional missing 10%	0.093	0.482	0.482	0.449	0.369	0.403	0.482
Optional missing 30%	0.075	0.476	0.476	0.428	0.372	0.402	0.476
Optional missing 50%	0.066	0.477	0.477	0.418	0.376	0.404	0.477
Edges removed 15%	0.110	0.469	0.469	0.411	0.362	0.386	0.469
Edges removed 30%	0.110	0.436	0.436	0.370	0.326	0.323	0.436
Typed rewire 15%	0.110	0.256	0.256	0.217	0.237	0.220	0.256
Typed rewire 30%	0.110	0.203	0.203	0.170	0.199	0.181	0.203
Lineage missing 30%	0.110	0.293	0.292	0.261	0.250	0.223	0.490

Condition	Coverage	Selective risk	Accepted errors
Clean	0.956	0.014	0.393
No anchors/support	0.996	0.026	0.750
Attribute noise 30%	0.930	0.014	0.413
Attribute noise 50%	0.926	0.014	0.427
Optional missing 10%	0.956	0.013	0.386
Optional missing 30%	0.956	0.013	0.371
Optional missing 50%	0.956	0.012	0.354
Edges removed 15%	0.957	0.014	0.398
Edges removed 30%	0.956	0.016	0.442
Typed rewire 15%	0.964	0.022	0.507
Typed rewire 30%	0.960	0.025	0.608
Lineage missing 30%	0.976	0.020	0.397

Typed rewiring retains entity kind, silo and split. Full visible lineage does not establish correctness; missing prefix tags can still count toward generic support. Perturbation logs are evaluator-only. Source graphs are copied for stress rather than mutating the native imports.

S13. WDC attribute-only robustness

The six conditions use frozen official pairs, estimators and policies. The first table averages all 27 official configurations and three seeds; overlapping cells are included descriptively. The second averages the nine fully unseen configurations and three seeds. Test pools repeat across sizes, so these are correlated summaries without pooled confidence intervals.

Candidate retrieval is not rerun under defects. Full machine-readable rows preserve hardness, size, unseen percentage, seed, condition, method and denominator for all 2,430 measurements.

All official configurations (including flagged overlaps)

Condition	Exact	Fuzzy	Attr	Emb	Attr+emb
clean	0.000	0.331	0.539	0.374	0.554
optional missingness 10	0.000	0.331	0.538	0.372	0.548
optional missingness 30	0.000	0.331	0.533	0.369	0.539
optional missingness 50	0.000	0.331	0.531	0.363	0.533
attribute noise 30	0.000	0.313	0.420	0.354	0.436
attribute noise 50	0.000	0.305	0.360	0.343	0.369

Fully unseen configurations only

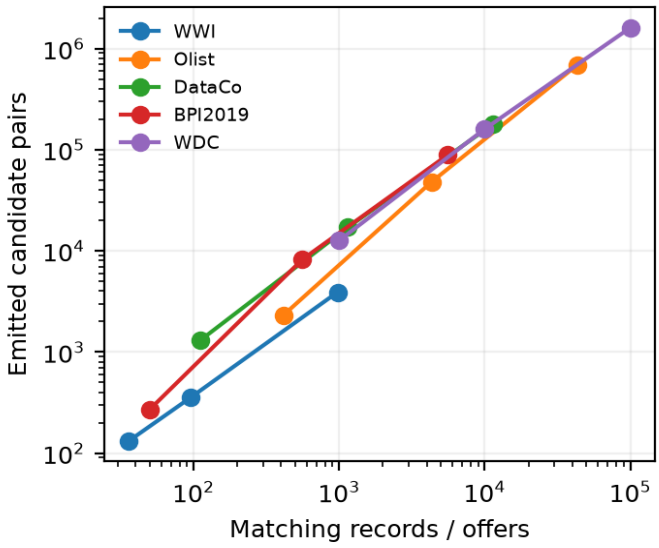
Condition	Exact	Fuzzy	Attr	Emb	Attr+emb
clean	0.000	0.348	0.537	0.392	0.558
optional missingness 10	0.000	0.348	0.534	0.389	0.551
optional missingness 30	0.000	0.348	0.528	0.386	0.542
optional missingness 50	0.000	0.348	0.528	0.382	0.534
attribute noise 30	0.000	0.326	0.419	0.370	0.439
attribute noise 50	0.001	0.315	0.359	0.355	0.371

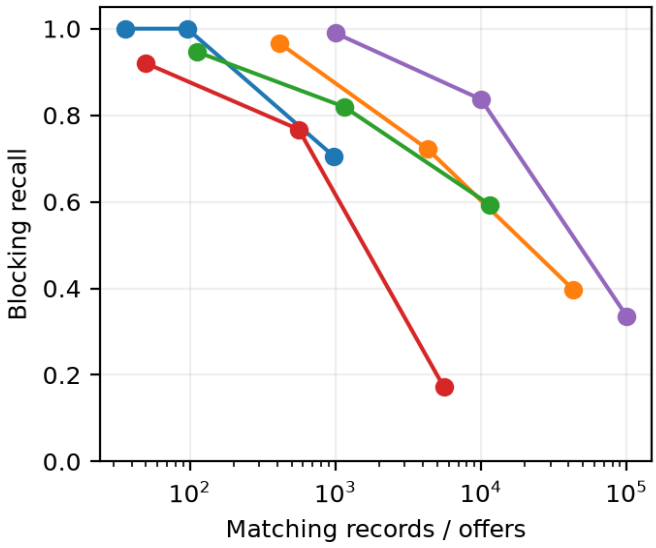
These offer attributes have no transaction or foreign-key graph. No graph-context, edge-noise or ERP-query result is fabricated for WDC.

S14. All main scaling points

Fifteen dataset/target points each have three trials, for 45 main trials. The table reports per-point medians. Actual is projected graph nodes for four datasets and offers for WDC. Complete graph units can exceed target. Matching records are shown separately. Later growth uses replicas with all cross-replica positives in the true-pair universe.

Dataset	Target	Actual	Match rec.	Pairs	Block R	Sec.	GiB
WWI	1k	1,021	36	132	1.000	0.25	0.32
WWI	10k	10,722	96	356	1.000	1.87	0.37
WWI	100k	100,729	974	3,896	0.705	24.46	0.70
Olist	1k	1,000	414	2,309	0.966	0.57	0.32
Olist	10k	10,001	4,308	47,913	0.722	7.56	0.48
Olist	100k	100,001	43,176	687,264	0.397	109.13	2.91
DataCo	1k	1,011	112	1,312	0.946	0.71	0.66
DataCo	10k	10,001	1,150	17,074	0.820	6.58	0.70
DataCo	100k	100,012	11,470	179,974	0.594	69.57	1.49
BPI2019	1k	1,032	50	270	0.920	0.50	1.08
BPI2019	10k	10,023	560	8,231	0.767	4.50	1.11
BPI2019	100k	100,014	5,566	88,716	0.172	47.62	1.49
WDC	1k	1,000	1,000	12,638	0.990	0.96	2.00
WDC	10k	10,000	10,000	160,137	0.837	18.74	2.11
WDC	100k	100,000	100,000	1,606,844	0.336	373.05	3.27





Emitted output is bounded by K per left record and a global guard. It does not prove subquadratic internal retrieval/ranking work. Falling recall precludes a recall-preserving scalability claim. Absolute RSS includes allocator retention; increments and every trial are recorded.

S15. Timing decomposition and query batches

Preparation, blocking, full features and all scorers are separate measured components. Main total additionally includes implemented query/load/timing work; it need not equal the sum of the selected components below. Median columns are aggregated independently across trials.

Dataset	Build s	Block s	Features s	Scorers s	Total s
WWI	1.768	0.220	9.706	0.033	24.464
Olist	1.934	8.270	66.777	10.919	109.131
DataCo	2.986	4.237	29.154	3.710	69.569
BPI2019	3.493	1.541	16.434	1.886	47.620
WDC	0.233	260.016	96.738	4.140	373.049

Dataset	Engine	p50 ms	p95 ms	Graph prep s	SQL prep s
BPI2019	graph	349.45	375.80	0.231	1.009
BPI2019	sql	1378.93	1393.95	0.231	1.009
DataCo	graph	415.09	476.25	0.234	0.883
DataCo	sql	1668.25	1826.90	0.234	0.883
Olist	graph	238.08	289.30	0.172	0.514
Olist	sql	1296.64	1341.63	0.172	0.514
WWI	graph	193.26	224.31	0.088	0.671
WWI	sql	768.46	841.87	0.088	0.671

Both engines use common per-copy oracle identities. Per-trial latencies have one warmup and five measured repetitions; the table shows median per-trial percentiles at the largest target. Full query_latencies.csv has all targets, templates and trials. No production optimizer or distributed-engine comparison is claimed.

Supplemental scorer_trials.csv has 135 isolated timings, and canonicalization_trials.csv has 36. Graph scorer targets are projected node volumes. WDC supplemental targets are 1k/10k/100k input pair rows, tiled from an official frozen feature matrix; they are not source-offer targets. Features are excluded and each scorer has one warmup plus three timed repetitions. These isolated costs do not give an end-to-end overhead ratio.

Hardware: Windows 11; Intel Family 6 Model 198; 24 logical CPUs; 68,137,205,760 bytes RAM (about 63.46 GiB); Python 3.12.10. Main experiment lock includes NumPy 2.4.3, pandas 2.3.3, scikit-learn 1.8.0, NetworkX 3.6.1 and ONNX Runtime 1.31.0. The manuscript uses a separate dependency environment and does not change the experiment lock.

S16. Reproduction, amendments and rights

Rebuild manuscripts. Install generated/manuscript_requirements.lock in a separate Python environment and install MiKTeX/TeX Live with IEEEtran, pdflatex and bibtex, plus Poppler. The source package rebuilds all PDFs from its generated tables, data and figures with scripts/build_fullstudy_paper.py. Regenerating figures/tables with scripts/prepare_fullstudy_paper.py and independently checking every source hash with scripts/verify_fullstudy_paper.py require the complete private experiment workspace, including inputs deliberately excluded from the manuscript package. Neither manuscript script refits models.

Verify experiments. The existing scripts/verify_all_datasets.py uses the recorded full-study environment to recompute metrics, replay frozen scores and execute fresh queries. It is a read-only verification of saved evidence. Reacquisition/model fitting requires the documented source assets and may take substantially longer. The expanded study has not undergone an independent fresh-environment retraining replication; the earlier primary study has.

Amendments. The interrupted WWI continuation preserved fitted models. Q1 SQL optimization deduplicated master/tag incidence before joining identities; all WWI E5 trials were rerun, preserving earlier six timings and source. Olist seller-state paths followed a test-context diagnostic before fitting. A later order-item lineage correction reran all Olist query controls and nine E5 trials while preserving models; exact frozen score replay verifies unchanged matching.

Integrity. claim_registry.json records SHA-256, aggregation rules and interpretation guards for generated claims. The manuscript verification manifest records PDF/source hashes, bibliography checks, numerical checks, page limits, rendering and visual review. The original four-page PDF is preserved. WDC files retain official pair lists and overlap annotations.

Responsible availability. No human study, recruitment or institutional approval is claimed. DataCo email/password fields are excluded from graph/model attributes; raw source references remain local. DataCo acquisition uses a documented mirror and does not establish publisher-byte equality. BPI publisher MD5 is checked. Olist license is CC BY-NC-SA 4.0; DataCo/BPI publisher metadata list CC BY 4.0; the frozen model card lists Apache 2.0. Microsoft sample and project release terms must be retained separately.

The private manuscript package excludes raw datasets, source SQL databases, evaluator truth and downloaded model weights. Authors must supply authorship, funding/conflicts, license choices and approval before external publication. No public artifact DOI, accepted paper, final camera-ready version or completed submission is claimed.

AI assistance. OpenAI Codex (GPT-6) assisted implementation, orchestration, checks, drafting all manuscript sections, bibliography organization and generation of tables/plots/publication files. Executed scripts and immutable measured artifacts supply numerical evidence. The final human authors remain responsible for reviewing methods, claims and declarations.

Sources. The detailed and IEEE papers share references.bib with primary research papers, dataset publisher records and official model documentation. Bibliography provenance records checked URLs and metadata. WDC official/disjoint rows, all calibration/robustness/latency trials and exact source hashes are available in the local artifact tree.

S17. Schema and descriptive decision cases

The schema preserves native typed records and relationships. Matching exposes a declared projection, while canonical entities are an added identity view. The examples below are selected after evaluation from saved accepted WWI decisions; they do not modify features, policies or labels. WWI names are fictional sample-data descriptions.

Layer	Nodes and relationships	Lineage / control
Native source	WWI Supplier/StockItem/PO/PO line/movement/finance; analogous commerce/process records in other adapters.	Source table and row; declared FK relationship sets checked independently.
Matching projection	Master -HAS_TRANSACTION-> Transaction -HAS_CATEGORY-> Category; WWI additionally typed master relations.	Local silo/split preserved; source-table/batch/mapper completeness; permitted typed summaries.
Decision view	Source -REFERS_TO-> CanonicalEntity; accepted cross-source links recorded as ACCEPTED_SAME_AS.	Only after frozen scoring; type and one-to-one constraints; rejected/abstained/conflicting decisions logged.
Hidden evaluator	Original entity truth and source-row reference answers.	Labels, splits, metrics and query oracles only; not text/context/blocking features.

Accepted true pair (seed 17)

IT joke mug - hardware: part of the computer that can be kicked (White)
IT joke mug - hardware: part of the computer that can be kicked (White)

Score 0.978238; frozen accept 0.897349; context used True; independent anchors 3; support 1.250. Candidate 12630161489751553ea9f3d2. Origins: Warehouse.StockItems / Warehouse.StockItems.

Accepted false merge (seed 29)

"The Gu" red shirt XML tag t-shirt (Black) S
"The Gu" red shirt XML tag t-shirt (Black) XXS

Score 0.992861; frozen accept 0.918069; context used True; independent anchors 4; support 2.000. Candidate f7de6934d1adf6423d7ecbcd. Origins: Warehouse.StockItems / Warehouse.StockItems.

These cases show recorded decision evidence, not causal feature attribution. Support or a high score does not certify identity. Exact model checksum and decision trigger are stored in decision_examples.json; evaluator truth remains excluded from the release package.